

DANI ROYTBURG

droytbur@andrew.cmu.edu | [djroytburg.github.io](https://github.com/droytburg) | [in/danijroytburg](https://in.danijroytburg)

Education

Carnegie Mellon University, School of Computer Science

Pittsburgh, PA

M.S., Machine Learning

expected May 2027

Relevant Coursework: Advanced Machine Learning, Probability and Statistics, Probabilistic Graphical Models, Machine Learning in Practice

Emory University, College of Arts and Science

Atlanta, GA

B.S., summa cum laude: Computer Science, Quantitative and Social Sciences.

August 2021-May 2025

Cumulative GPA: 3.85/4.0 | **Dean's List** Spring 2024

Relevant Coursework: Deep Learning, Machine Learning, Privacy-Enhanced Machine Learning, Systems Programming, Computer Architectures, Linear Algebra, Adv. Calculus, Probability and Statistics (I, II), Discrete Math, Algorithmic Analysis.

Work Experience

DI Group, Carnegie Mellon University

Pittsburgh, PA

Research Assistant, advised by [Daphne Ippolito](#)

September 2025 – present

- Develop multi-agent adversarial reinforcement learning environments to improve legibility of reasoning traces.

Martian Research

San Francisco, CA

Research Fellow, advised by [Jacob Haimes](#), [Phillip Quirke](#), Narmeen Oozeer

May 2025 – present

- Engineer activation profiles of language model evaluators to suppress self-preferential bias, applied to LLM routers.
- Winner of [Mechanistic Router Interpretability](#) Hackathon (40 submissions).

Digital Humanities Lab, Emory University

Atlanta, GA

Research Assistant, advised by [Lauren Klein](#) and [Sandeep Soni](#)

November 2021 – May 2025

- Designed and trained terabyte-scale graph-text models of linguistic network propagation (ICWSM '25).
- Evaluated adversarial robustness of large language models' commonsense reasoning to distributional shift.
- Built historical correspondence networks and analytic dashboards in (D3, Next.js), supported by PBS.

ZS Associates

Evanston, IL

Intern – Business Technology Solutions Associate

June 2024 – August 2024

- Automated ETL pipelines from diverse data sources for consumption forecasts of specialty neurological treatments.
- Implemented lossless restoration of compromised client data using synthetic data imputation techniques.

Cloverpop

Chicago, IL

Intern – Machine Learning

January 2024 – May 2024

- Deployed a decision management database with structure prediction on 65k meeting transcripts.
- Forecasted demand growth with traditional machine learning methods (decision trees, regression models).

McCormick and Company

Hunt Valley, MD

Intern – Global Marketing and Sales Data Science

June 2023 – August 2023

- Built self-sustaining Amazon sales data visualization, oversight, and reporting tools in Microsoft Power BI.
- Automated customer review database analytics with GCP BigQuery, augmented with BERT-based embedding search.

Select Publications

[Breaking the Mirror](#): Breaking the Mirror: Activation-Based Mitigation of Self-Preference in LLM Evaluators (2025)

Dani Roytburg, Narmeen Oozeer et. al, *Mechanistic Interpretability, Eval-LLM Workshops @ NeurIPS*

[Mind the Gap](#): Pathways Towards Unifying AI Safety and Ethics Research (2026)

Dani Roytburg, Beck Miller, *International Association for Safe & Ethical AI (IASEAI)*

[Words and Action](#): Modeling Linguistic Leadership in #BlackLivesMatter (2025)

Dani Roytburg, Debbie Olorunsola, Sandeep Soni, Lauren Klein, *International AAAI Conference on Web and Social Media*

Technical Skills

Languages: Python, Java, R, JavaScript, Typescript, SQL || French (C1 proficiency), Russian (C1), Chinese (B1)

Packages: PyTorch, JAX, HuggingFace Transformers, scikit-learn, multiprocessing, spaCy, networkx, D3.js, React.js

Databases and Deployment: Firebase, Vercel, Amazon EMR, Google Cloud Platform (GCP), Docker, MySQL

Dev Tools: Git, Linux, Power BI, Tableau, Figma